Arif Hassan Zidan

Doctoral Student, School of Computer & Cyber Sciences, Augusta University

+1 (706) 619-7098 arifhassanzidan@gmail.com LinkedIn

Google Scholar Profile Homepage



Skills

Languages	Strong reading, writing and speaking competencies for English and Bengali.
Coding	Python, Java, PHP, SQL, MATLAB, Latex.
Databases	MySQL, PostgreSQL.
Web Dev	HTML, CSS, JavaScript, TypeScript, SAPUI5.

Research Interests

- Artificial Intelligence; Machine Learning; Multimodal Large Language Models; Computational Neuroscience; Brain Imaging; Quantum Machine Learning.

Work & Research Experience

August 2024 – present

Graduate Research Assistant

Augusta University, Augusta, Georgia, USA. *Supervisor: Dr. Wei Zhang*

Research Focus: Artificial intelligence and deep learning methods for brain imaging analysis and neural signal reconstruction using multimodal data. Implementing Quantum Machine Learning Algorithms for educational measurement data.

June 2026 – August 2026

Internship

Department of Radiology, Massachusetts General Hospital, Harvard Medical School, Boston, MA, USA. *Advisor: Dr. Xiang Li*

As a student intern at the Center for Advanced Medical Computing and Analysis (CAMCA), engineered robust data pipelines using Python to process large-scale multimodal medical imaging datasets, and assisted with algorithm development, model deployment, and validation under faculty and clinical supervision. Actively participated in interdisciplinary meetings, documented technical findings, and strictly adhered to the ethical, privacy, and regulatory standards required at MGH.

September 2022 – June 2024

SCT Bangla Limited, Bangladesh, Junior Software Developer.

- Developed and maintained full-stack ERP (Enterprise Resource Planning) software using SAPUI5, TypeScript, and JavaScript.
- Designed database tables and wrote efficient queries using PostgreSQL to manage backend enterprise data.
- Managed code versions and collaborated with the development team using Git and Bitbucket.
- Wrote clean, functional code across the full software development lifecycle to support reliable daily operations.

February 2021 – March 2021

Bangladesh Telecommunications Company Limited, Internship.

Education

2024 – Present
Ph.D., Augusta University, Computer & Cyber Sciences Ph.D. program.
CGPA: 4.00 (current)
Anticipated Graduation Date: December 2029
Relevant Coursework: Algorithm Analysis, Machine Learning, Human-computer Interaction, Data Science, Network & Dist. Algorithms.

2018 – 2022
B.Sc.(Hons), Bangladesh University of Professionals, Bangladesh in Information and Communication Engineering.
CGPA: 3.86/4.00
Thesis title: *Case Fatality and Intubation Requirement Prediction of COVID-19 Patients Affected with Various Severe Diseases Using Machine Learning Classification Algorithms*.
Relevant Coursework: Data Structure, Analysis and Design of Algorithm, Object Oriented design and Programming, Discrete Mathematics, Operating System, Database Management Systems, Data Communication & Computer Networks, Cloud Computing, Software & Requirements Engineering, System Analysis & Design, Software Testing and Maintenance, Artificial Intelligence, Machine Learning.

2017
Higher Secondary School Certificate, Notre Dame College, Bangladesh
GPA: 5.00/5.00.

Research Publications

Journal Articles

1. **A. H. Zidan**, A. Jahin, Y. Bao, T. Liu, and W. Zhang, "A theoretical and experimental exploration in permutation randomization on nonsmooth nonconvex optimization," *Contemporary Mathematics*, pp. 2382–2403, Apr. 2026, ISSN: 2705-1064. DOI: 10.37256/cm.7220268464
2. **A. H. Zidan**, A. Jahin, Y. Bao, and W. Zhang, "Semi-nonlinear deep efficient reconstruction for unveiling linear and nonlinear spatial features of the human brain," *Electronics (2079-9292)*, vol. 14, no. 3, 2025.
3. **A. H. Zidan**, A. Jahin, Y. Bao, and W. Zhang, "Mapping longitudinally consistent intrinsic connectivity networks in macaque brain via longitudinal sparse dictionary learning," *IBRO Neuroscience Reports*, 2024, ISSN: 2667-2421. DOI: https://doi.org/10.1016/j.ibneur.2024.11.014
4. **A. H. Zidan** and M. M. Islam, "Case fatality and intubation prediction of covid-19 patients affected with severe diseases using machine learning classifiers," *BUP Journal*, vol. 10, no. 1, 2023.

Conference Proceedings

1. T. Ahmed, M. H. Amin, M. S. Uddin, B. Uddin, N. J. Ema, R. Hassan, and **A. H. Zidan**, "Boundary-aware deep learning framework for breast cancer image segmentation: Enhancing tumor delineation through biou-driven evaluation," in *2026 IEEE 18th International Conference on Computational Intelligence and Communication Networks (CICN)*, IEEE, 2026, pp. 296–301. DOI: 10.1109/cicn70047.2026.11594147
2. M. S. Uddin, M. H. Amin, N. J. Ema, B. Uddin, T. Ahmed, and **A. H. Zidan**, "Multilingual financial fraud detection using machine learning and transformer models: A bangla-english study," in *2026 IEEE 18th International Conference on Computational Intelligence and Communication Networks (CICN)*, IEEE, 2026, pp. 495–500. DOI: 10.1109/cicn70047.2026.11594250
3. **A. H. Zidan**, Y. Bao, T. Liu, and W. Zhang, "Decoding reproducible functional connectivity networks in the frequency domain with deep learning and large language models," IEEE EMBC, 2026.

4 **A. H. Zidan**, M. Fouda, Z. M. Fadlullah, and M. I. Ibrahem, "A systematic survey on cryptography-driven privacy in machine learning," in *2026 IEEE 2nd International Conference on Secure IoT, Assured and Trusted Computing (SATC)*, IEEE, 2026, pp. 1–5. DOI: 10.1109/satc69565.2026.11542242

5 **A. H. Zidan**, B. Uddin, N. J. Ema, M. H. Amin, M. S. Uddin, and T. Ahmed, "Evaluating parameter-efficient fine-tuning of open-weight llms for medical question answering," in *2026 IEEE 18th International Conference on Computational Intelligence and Communication Networks (CICN)*, IEEE, 2026, pp. 374–379. DOI: 10.1109/cicn70047.2026.11594337

Full list of 17+ publications available on: [Google Scholar].

Miscellaneous Experience

Awards and Achievements

2026 **Student Spotlights**, The Graduate School, Augusta University. Read the full article here.

Featured in "Building Community and Leadership: The Graduate Ambassador Experience", JagWire, Augusta University. Read the full article here.

2025 **Graduate School Excellence Award at Scientific Integrity Day**, Augusta University.

2020 **BUP Merit Scholarship**, Bangladesh University of Professionals (BUP).

2019 **Dean's Appreciation Award**, Bangladesh University of Professionals (BUP).

Oral Presentation

2026 **Decoding Reproducible Functional Connectivity Networks in the Frequency Domain with Deep Learning and Large Language Models**
IEEE EMBC 2026 Conference, Toronto, Canada
Selected for oral presentation, a distinction awarded to approximately 10–15% of accepted submissions.

Poster Presentation

2024 **Revolutionizing Neurological Disorders Diagnostics: The Power of Large Language Models and Prompt Tuning in fMRI Analysis**
Organized by: Computer & Cyber Graduate Student Organization, Augusta University

Conference Reviewer

MICCAI 2025, MICCAI 2026, MIUA 2026.

Extracurricular Activities

August 2026 - July 2027 **Student Representative**, Auxiliary Services Division, Augusta University.

August 2026 **Volunteer**, TechNet Augusta 2026.

September 2025 - August 2026 **Graduate Ambassador**, The Graduate School, Augusta University.

August 2025 - present **Representative**, Graduate Student Government Association, Augusta University.

Miscellaneous Experience (continued)

January 2025 - December 2025 — **Cultural Secretary**, Augusta University Bangladeshi Students Association (AUBSA).